

1 Rebuttal 1-page PDF (Global Response)

Token	Metric	Rnd-Words	Noise-Imgs	Simple	Semi-NMF (Ours)	GT-captions
Dog	CS top-1 ($\uparrow$)	0.537 ± 0.03	0.530 ± 0.05	<u>0.567 ± 0.08</u>	0.595 ± 0.07	0.777 ± 0.06
	BS top-1 ($\uparrow$)	0.205 ± 0.07	0.227 ± 0.06	0.331 ± 0.07	<u>0.305 ± 0.07</u>	0.519 ± 0.11
Bus	CS top-1 ($\uparrow$)	0.509 ± 0.04	0.487 ± 0.05	0.619 ± 0.06	<u>0.591 ± 0.08</u>	0.742 ± 0.05
	BS top-1 ($\uparrow$)	0.198 ± 0.07	0.253 ± 0.06	0.319 ± 0.04	<u>0.306 ± 0.06</u>	0.460 ± 0.10
Train	CS top-1 ($\uparrow$)	0.518 ± 0.03	0.505 ± 0.04	<u>0.633 ± 0.05</u>	0.640 ± 0.07	0.725 ± 0.05
	BS top-1 ($\uparrow$)	0.177 ± 0.07	0.221 ± 0.04	0.310 ± 0.05	<u>0.293 ± 0.05</u>	0.432 ± 0.08
Cat	CS top-1 ($\uparrow$)	0.536 ± 0.03	0.545 ± 0.04	0.625 ± 0.06	<u>0.621 ± 0.07</u>	0.795 ± 0.05
	BS top-1 ($\uparrow$)	0.142 ± 0.06	0.235 ± 0.05	<u>0.306 ± 0.06</u>	0.329 ± 0.07	0.540 ± 0.11

Table 1: Concept extraction on LLaVA-v1.5: Test data mean CLIPScore reported for top-1 activating concept for same baselines and tokens as in main paper table 1. Higher scores are better. Best score in **bold**, second best is underlined.

Token	Simple	PCA	KMeans	Semi-NMF
Dog	0.435	0.008	0.429	<u>0.149</u>
Bus	0.464	0.010	0.518	<u>0.124</u>
Train	0.315	0.024	0.382	<u>0.087</u>
Cat	0.479	0.013	0.554	<u>0.166</u>

Table 2: Overlap evaluation (LLaVA). Lower is better. Best score in **bold**, second best underlined.



Figure 1: Multimodal grounding for 6 out of 20 concepts for 'Dog' token (layer 31) on LLaVA.

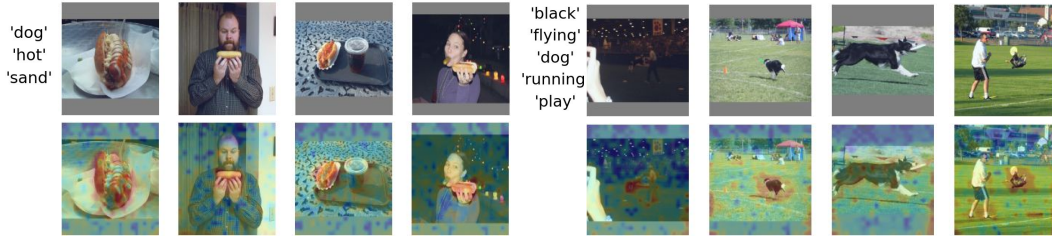


Figure 2: Saliency map for concept activation for two concepts on LLaVA ('Dog' token). First row shows the multimodal grounding. Second row shows the generated saliency map to indicate concept activation. Red denotes region of high activation and blue denotes region of low activation.